

Introduction to Programming

Key Points

- Programming is about giving instructions to a computer to perform a specific task.
- A computer program is a set of instructions that a computer can execute.
- To write a program, we need to understand the problem statement, identify the inputs, and determine the desired output.
- We use a programming language to communicate with the computer.
- **Algorithms** and **flowcharts** are used to represent the steps involved in solving a problem.

Understanding the Problem Statement

- To solve a problem, we need to understand what the problem is asking for.
- We need to identify the inputs, processing, and output required to solve the problem.
- For example, if we want to make tea, we need to know the ingredients, the steps involved, and the expected output (a cup of tea).
- **Input**, **processing**, and **output** are the key components of a problem statement.

Breaking Down the Problem

- We break down the problem into smaller, manageable steps.
- We identify the inputs, processing, and output required for each step.
- For example, to make tea, we need to:
 - Boil water
 - Add tea leaves
 - Add sugar
 - Add milk
- Each step involves a specific input, processing, and output.

Creating a Flowchart

- A flowchart is a graphical representation of the steps involved in solving a problem.
- It consists of **parallelograms** (input/output), **rectangles** (processing), and **diamonds** (decision-making).
- We use flowcharts to visualize the steps involved in solving a problem and to identify the inputs, processing, and output required.

Key Takeaways

Study Notes

- Programming involves giving instructions to a computer to perform a specific task.
- We need to understand the problem statement, identify the inputs, and determine the desired output.
- **Algorithms** and **flowcharts** are essential tools for solving problems and writing programs.
- Breaking down a problem into smaller steps and identifying the inputs, processing, and output required for each step is crucial for writing an effective program.